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MAY/JUNE 2007

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

CHEMISTRY

Paper 03 – General Proficiency

1 hour

23 MAY 2007 (a.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **In addition to the 1 hour, candidates are allowed a reading time of 10 minutes. Writing may begin during the 10-minute period.**
2. **Answer THREE questions on this paper, ONE from Section A, ONE from Section B and ONE from Section C.**
3. **All working MUST be shown for calculations.**
4. **The use of non-programmable calculators is allowed.**

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SECTION A

Answer ONE question from this section.

1. (a) Phosphoric acid (H_3PO_4), which is used as an additive in soft drinks, can react with sodium hydroxide to form both normal and acidic salts.

(i) Define the terms 'acid salts' and 'normal salts'. **(2 marks)**

(ii) Write ONE balanced equation EACH for the formation of a normal salt and an acid salt when phosphoric acid reacts with aqueous sodium hydroxide. **(4 marks)**

(iii) Excess consumption of soft drinks may lead to excess stomach acid.

a) How is this problem usually treated?

b) Explain the chemical principles upon which the treatment you stated in (iii) a) above is based.

c) Write an ionic equation for the reaction.

(4 marks)

(b) 25.0 cm^3 of 2.0 mol dm^{-3} sodium hydroxide and 50.0 cm^3 of 1.0 mol dm^{-3} hydrochloric acid, both at 30.0°C, are mixed together in a styrofoam cup. On mixing, the temperature rises to 38.9°C.

(i) Calculate the heat of neutralisation for the reaction in kJ mol^{-1}.

Use the following information in your calculation:

- Heat evolved = mass (g) x specific heat capacity, C, (Jg^{-1} $°C^{-1}$) x change in temperature
- C = 4.2 Jg^{-1} $°C^{-1}$
- Density of solution = 1 g/cm^3 **(5 marks)**

(ii) Draw a labelled energy profile diagram to illustrate the reaction in (b) (i) above. **(3 marks)**

(c) A solution of phosphoric acid has a pH of 4 and a solution of hydrochloric acid has a pH of 1. Which of the two is the stronger acid? Explain your answer. **(2 marks)**

Total 20 marks

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2. In Experiments I and II, 6.5 g of zinc granules are allowed to react with excess 1 mol dm^{-3} sulphuric acid at 25°C and 80°C respectively. Figure 1 shows the rate of production of hydrogen gas for both experiments.

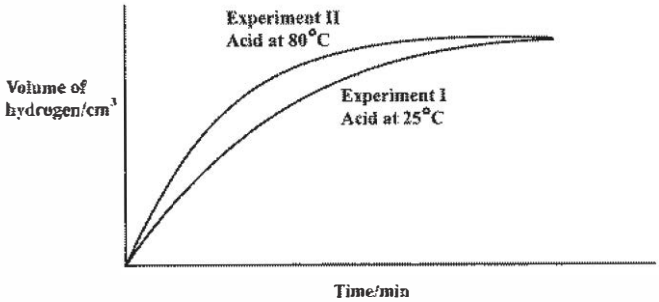

Figure 1. Production of hydrogen from zinc and sulphuric acid

(a) (i) Write a balanced equation for the reaction between zinc and aqueous sulphuric acid. **(2 marks)**

(ii) Based on the balanced equation you wrote in (a) (i) above, calculate the maximum volume of hydrogen gas at r.t.p. which can be generated from Experiment I using 6.5 g of zinc granules. **(2 marks)**

(R.A.M: H = 1, Zn = 65, S = 32, O = 16; 1 mole of gas occupies 24 dm^3 at r.t.p.)

(b) In a third experiment, Experiment III, 6.5 g of zinc reacts with excess 2 mol dm^{-3} sulphuric acid at 80°C.

(i) Make a sketch of the graph in Figure 1 in your answer booklet. On your sketch, draw the expected results for Experiment III. **(2 marks)**

(ii) With reference to the slopes of the graphs for Experiments I, II and III, discuss the effects of EACH of the following on the rate of production of hydrogen:

a) Temperature

b) Concentration of the acid **(8 marks)**

(c) State TWO precautions that should be taken when conducting these experiments. **(2 marks)**

(d) Catalysts are used to alter the rates of chemical reactions.

(i) Explain how a catalyst works to increase the rates of chemical reactions. **(2 marks)**

(ii) Draw a labelled energy profile diagram to show how a catalyst would alter the rate of reaction between zinc and sulphuric acid. **(2 marks)**

Total 20 marks

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SECTION B

Answer ONE question from this section.

3. Carefully study the reaction scheme in Figure 2 and answer the questions which follow.

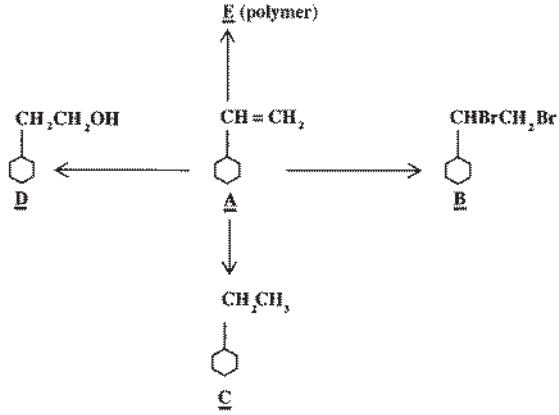


Figure 2. Reaction scheme

(a) Identify suitable reagents and reaction conditions which can be used to convert

(i) Compound A into Compound B

(ii) Compound A into Compound C.

(4 marks)

(b) Describe what may be OBSERVED when Compound D reacts with EACH of the following:

(i) Acidified potassium chromate (VI)

Give a reason for your answer. **(3 marks)**

(ii) Sodium metal

Include a balanced chemical equation in your answer. **(3 marks)**

(c) Table 1 below gives TWO statements about Compound A. These statements may be true or false.

Copy this table in your answer booklet and fill in the information required in the blank spaces. In answering this question you should refer to the information about Compouud A given in Figure 2 on page 4.

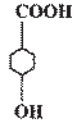
TABLE 1

Statements about Compound A	**Indicate whether true or false**	**Reason**	**Where statement is false, give correct statement**
(i) Compouud A is a saturated hydrocarbon.			
(ii) Compound A undergoes condensation polymerization.			

(6 marks)

(d) (i) Draw the partial structure of the polymer, E, that is formed when Compound A is polymerized. **(2 marks)**

(ii) State how the structure of the polymer you have drawn in (d) (i) above differs from that formed from the polymerization of Compound F shown below. Illustrate your answer by means of an appropriate partial structure.



Compound F

(2 marks)

Total 20 marks

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4. (a) (i) Show by means of chemical equations, the main steps involved in the manufacture of sulphuric acid by the Contact process, starting with sulphur. You should state the reaction conditions required. **(6 marks)**

(ii) The manufacture of sulphuric acid is a highly exothermic process. Suggest how this helps to make the whole process more economical. **(1 mark)**

(iii) Suggest why in the manufacture of sulphuric acid, the sulphur trioxide is dissolved in sulphuric acid (98%) rather than directly in water. **(1 mark)**

(iv) Name TWO factors that should be taken into consideration in the siting of a sulphuric acid plant. **(2 marks)**

(b) Account for the following observation:

Sulphur burns in air to form an oxide which is gaseous, whereas sodium forms an oxide which is a solid, although both elements belong to the same period of the periodic table. **(4 marks)**

(c) Natural rubber is very soft and pliable. It can be strengthened by the process of vulcanization. Illustrate by means of a labelled diagram the role of sulphur in the vulcanization process. **(2 marks)**

(d) Consider the following equations and determine whether sulphuric acid and sulphur dioxide are acting as reducing or oxidizing agents. You should indicate how you arrived at your answer.

(i) $Cu(s) + 2H_2SO_4(aq) \rightarrow CuSO_4(aq) + SO_2(g) + 2H_2O(l)$

(ii) $SO_2(g) + 2H_2O(l) + Cl_2(g) \rightarrow H_2SO_4(aq) + 2HCl(aq)$

(4 marks)

Total 20 marks

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SECTION C

Answer ONE question from this section.

5. The increase in the intensity and frequency of hurricanes and flooding in the Caribbean has been attributed in some quarters to an increase in the average global temperature. The rise in global temperature has been partially attributed to the increase in greenhouse gases in the atmosphere.

(a) Briefly outline how greenhouse gases can contribute to an increase of the earth's atmospheric temperature.

Include in your answer:

(i) The name of the main greenhouse gas **(1 mark)**

(ii) TWO main causes of the increase in concentration of the named greenhouse gas **(2 marks)**

(iii) An explanation of how EACH of the causes named in (ii) above can lead to the increase of this gas in the atmosphere **(3 marks)**

You should include ONE chemical equation in your answer.

(iv) A brief explanation of how the greenhouse effect works. Use a labelled diagram to illustrate your answer. **(5 marks)**

(b) There is a claim that the use of electrical cars can completely eliminate the greenhouse gas effect. Do you support this statement? Suggest TWO reasons for your answer. **(2 marks)**

(c) Over the years, studies have shown that there has been a significant decrease in the levels of the ozone layer. This has been linked to the extensive use of chlorofluorocarbons (CFCs) for many years.

Discuss the significance of the above statement, making reference to:

- The function of the ozone layer
- The action of CFCs on the ozone layer
- TWO health or environmental effects of the reduction of the ozone layer
- The possible long-term implications for the Caribbean's tourism industry
- ONE measure that has been put in place to reduce the amount of CFCs in the atmosphere **(7 marks)**

Total 20 marks

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6. Figure 3 below shows a generalized structure of a detergent.

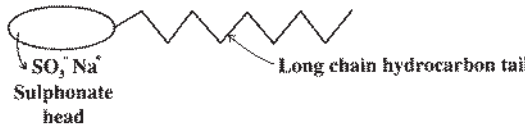

Figure 3. A generalised structure of a detergent

(a) Using the information given in Figure 3 above, suggest how detergents remove dirt from a piece of clothing during washing. **(4 marks)**

(b) In those areas that use hard water, when soaps are used for bathing, bathrooms must be cleaned very often. However, when detergents such as bath gels are used this frequent cleaning is not found to be necessary.

Suggest a reason for this. **(4 marks)**

(c) One major function of laundering clothes is stain removal.

Stacy-Ann has two similar pieces of cloth. On one piece there is a tea stain and on the other a rust stain.

(i) Suggest an appropriate reagent to effectively remove EACH stain. **(2 marks)**

(ii) Give ONE reason for the selection of EACH stain remover and explain how EACH of them works. **(4 marks)**

(d) Fabric softeners are usually added to the final rinse when laundering.

(i) What type of compounds are fabric softeners? **(1 mark)**

(ii) State THREE reasons why fabric softeners are used. **(3 marks)**

(iii) For ONE of the reasons identified in (d) (ii) above, explain how fabric softeners help to carry out this function. **(2 marks)**

Total 20 marks

END OF TEST